A Tour of the AI Labour Market

AI Job Market Analysis: Descriptive, Diagnostic, Prescriptive & Predictive Insights



Project Overview

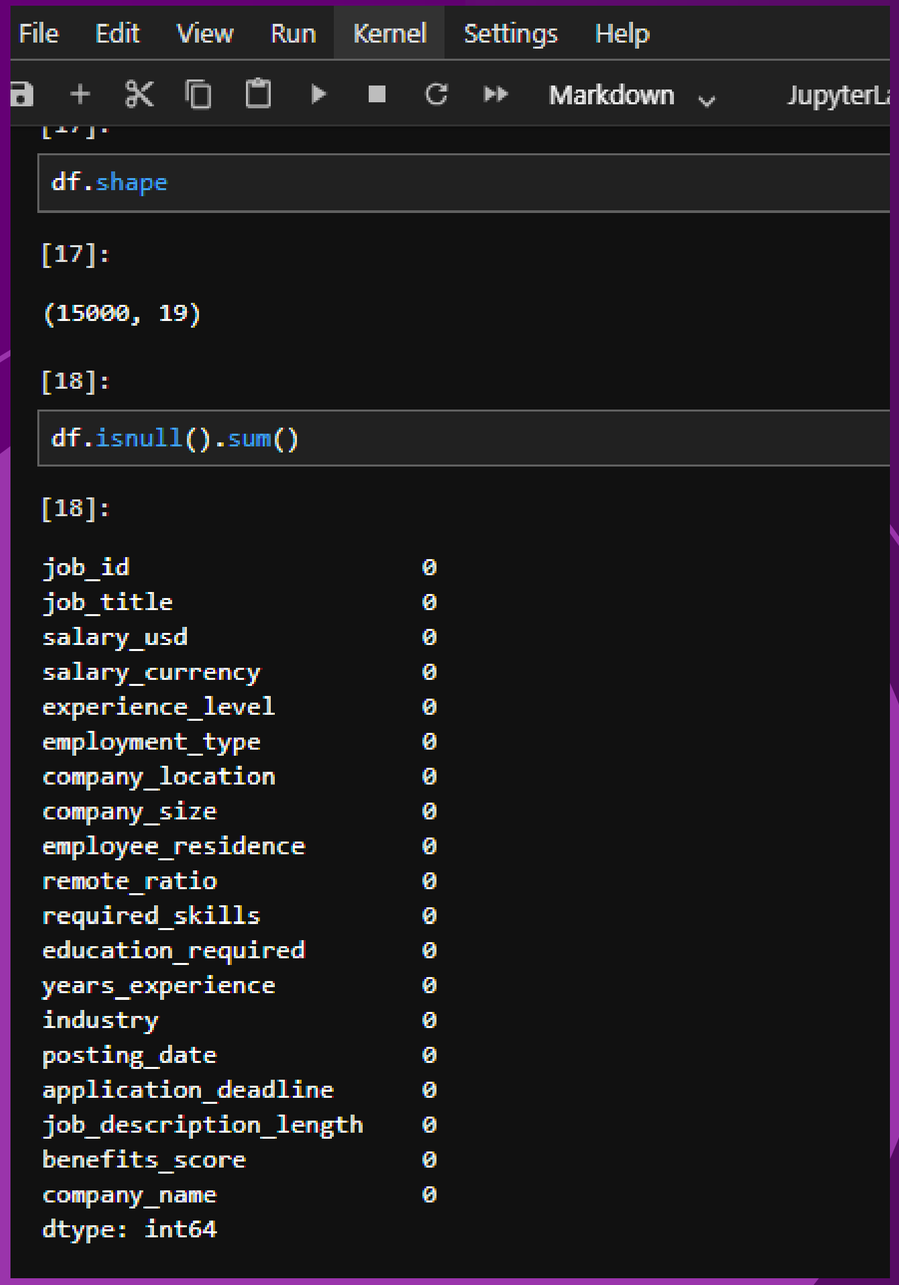
- Dataset obtained from Kaggle
- 15,000 rows and 19 columns
- Data was cleaned in Python to ensure readiness for analysis
- Type of analytics performed include; Descriptive, Diagnostic, Prescriptive and Predictive Analytics





Objectives of the Project

1. Analyze distribution of salaries across experience, education, and company size.
2. Identify top-paying industries and job roles.
3. Examine how benefits score, years of experience, and remote ratio relate to salaries.
4. Explore most in-demand skills.
5. Visualize geographical trends in AI job salaries.



Dataset Overview & Cleaning

- Checked for missing values and duplicates
- Cleaned text (standardized job titles and skills)
- Standardized data types across columns

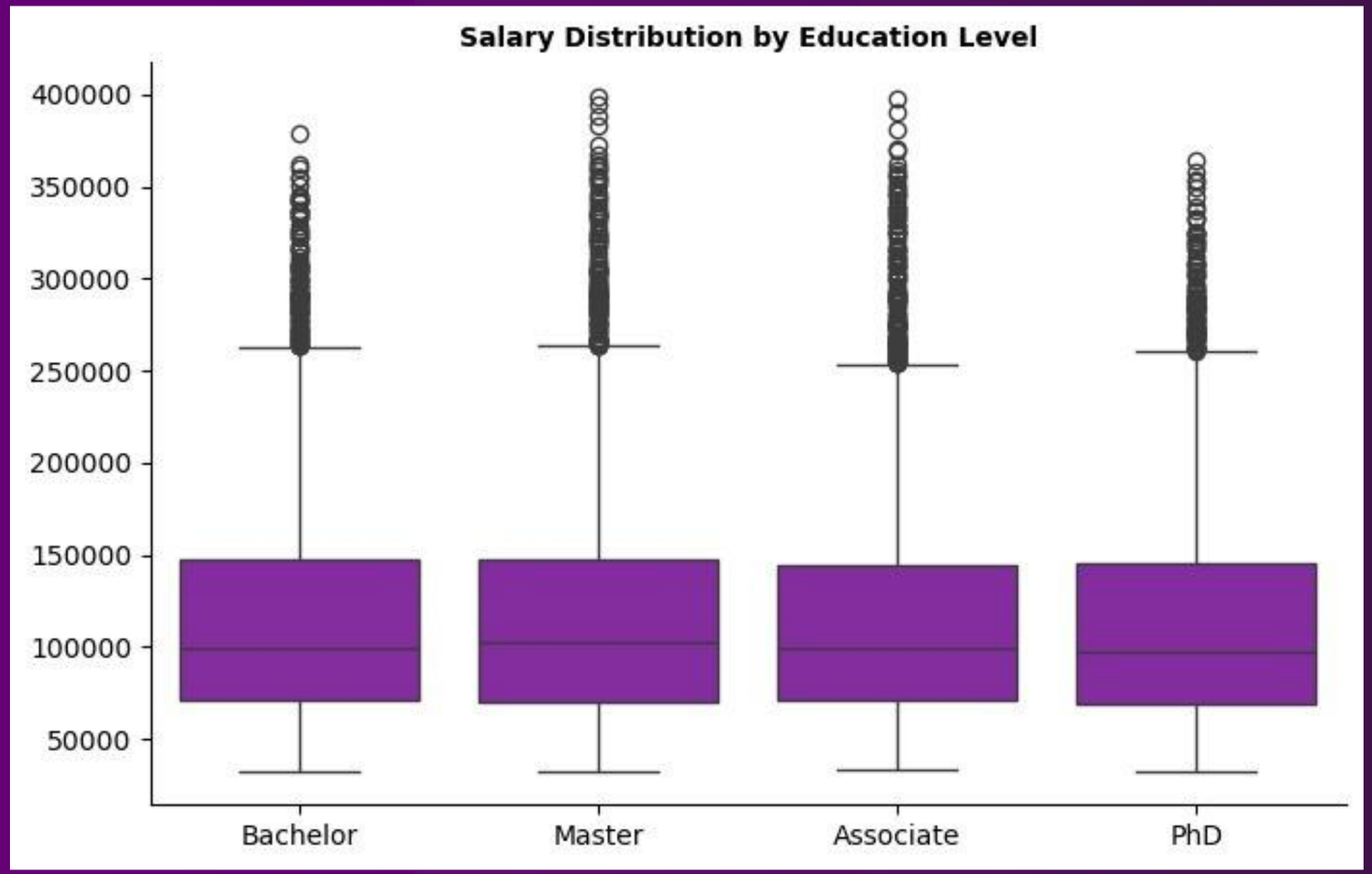
Descriptive and Diagnostic Analytics

Objective 1: Salary Distribution by Experience, Education, and Company Size



Senior-level (SE and EX) roles have the highest median salaries.

Objective 1: Salary Distribution by Experience, Education, and Company Size



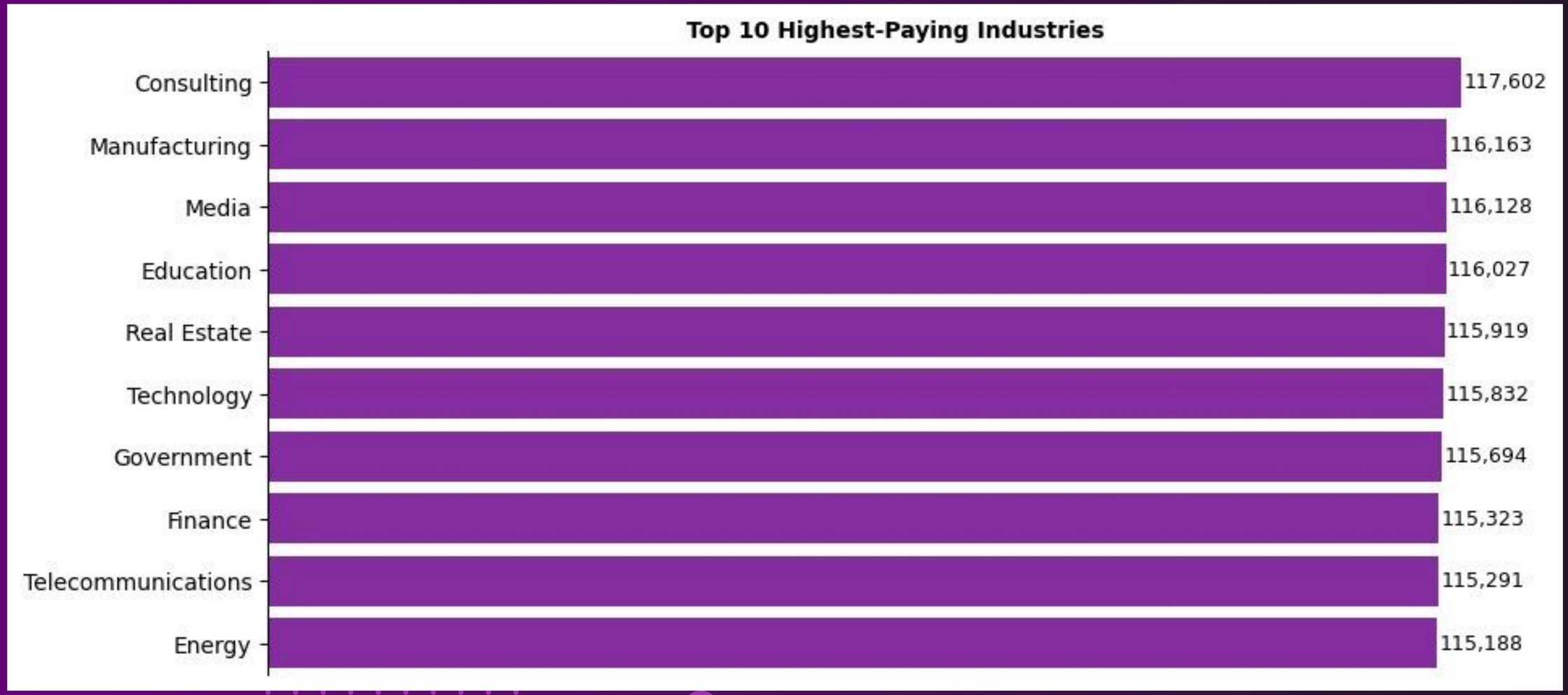
PhD-level roles show wider salary variation.

Objective 1: Salary Distribution by Experience, Education, and Company Size



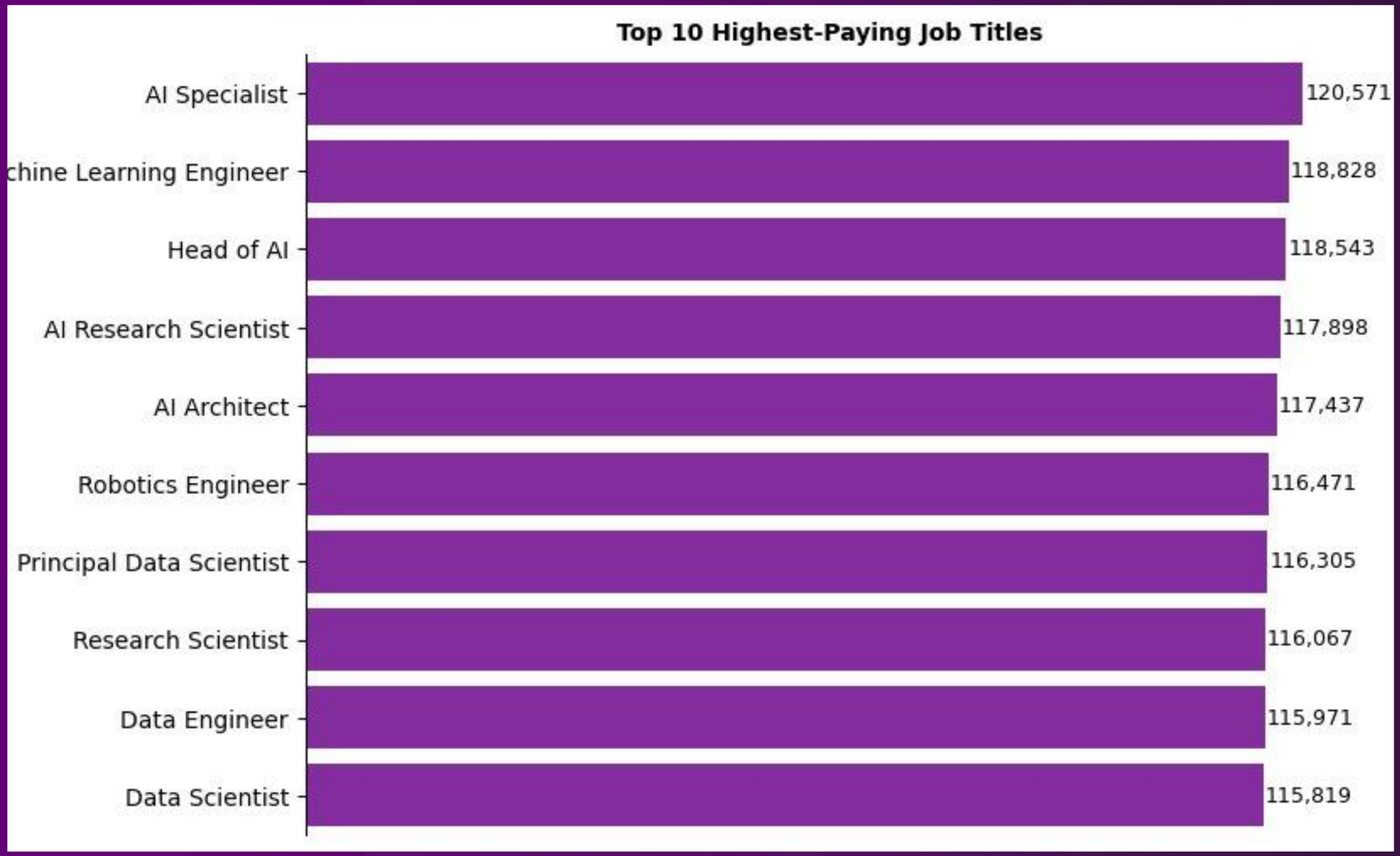
Larger companies tend to pay more consistently

Objective 2: Top Paying Industries and Roles



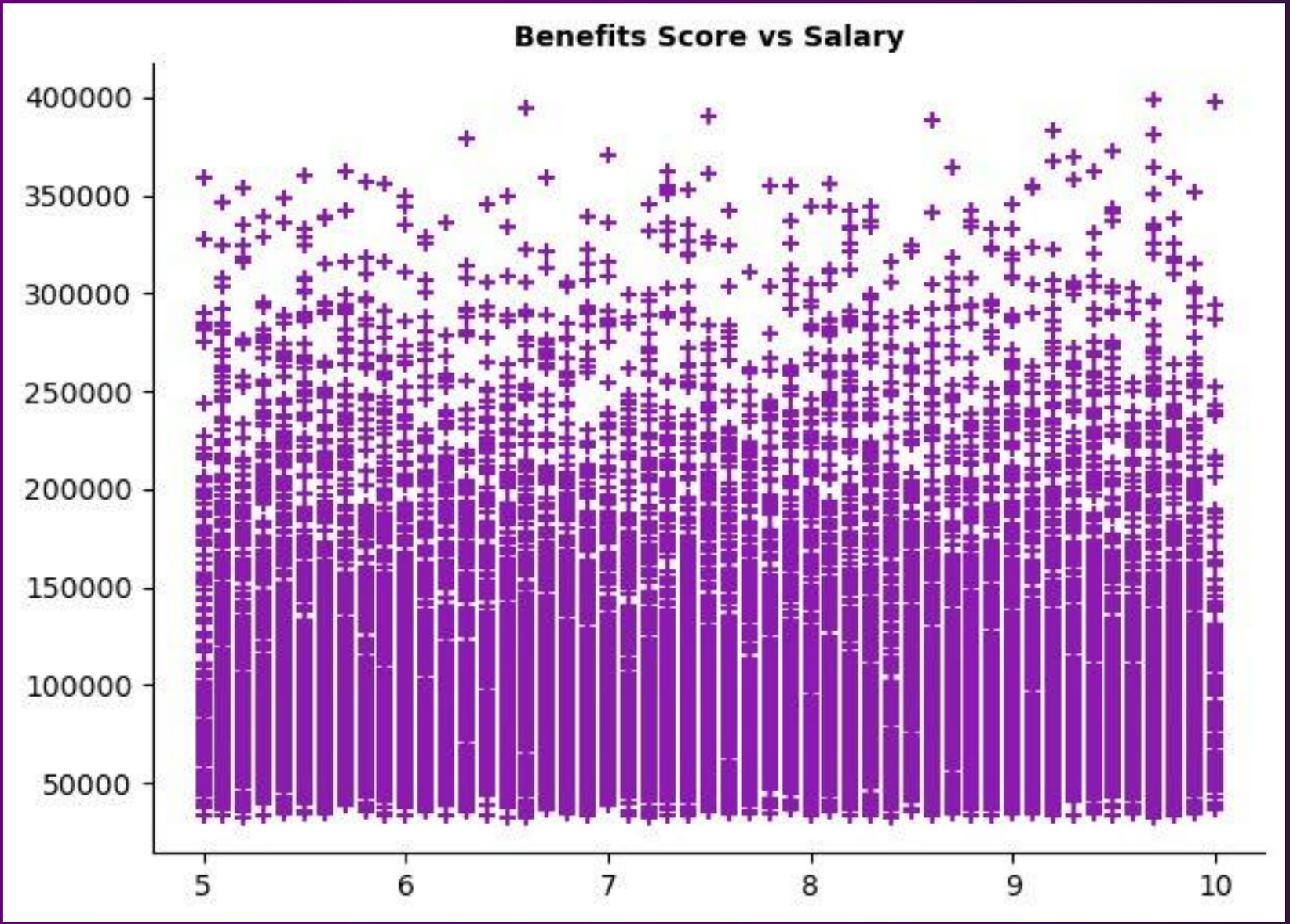
AI Research (Consulting) and Cloud Computing (Manufacturing) roles lead the salary charts

Objective 2: Top Paying Industries and Roles



AI Specialists and ML Engineers earn top-tier pay

Objective 3: Salary Relationships



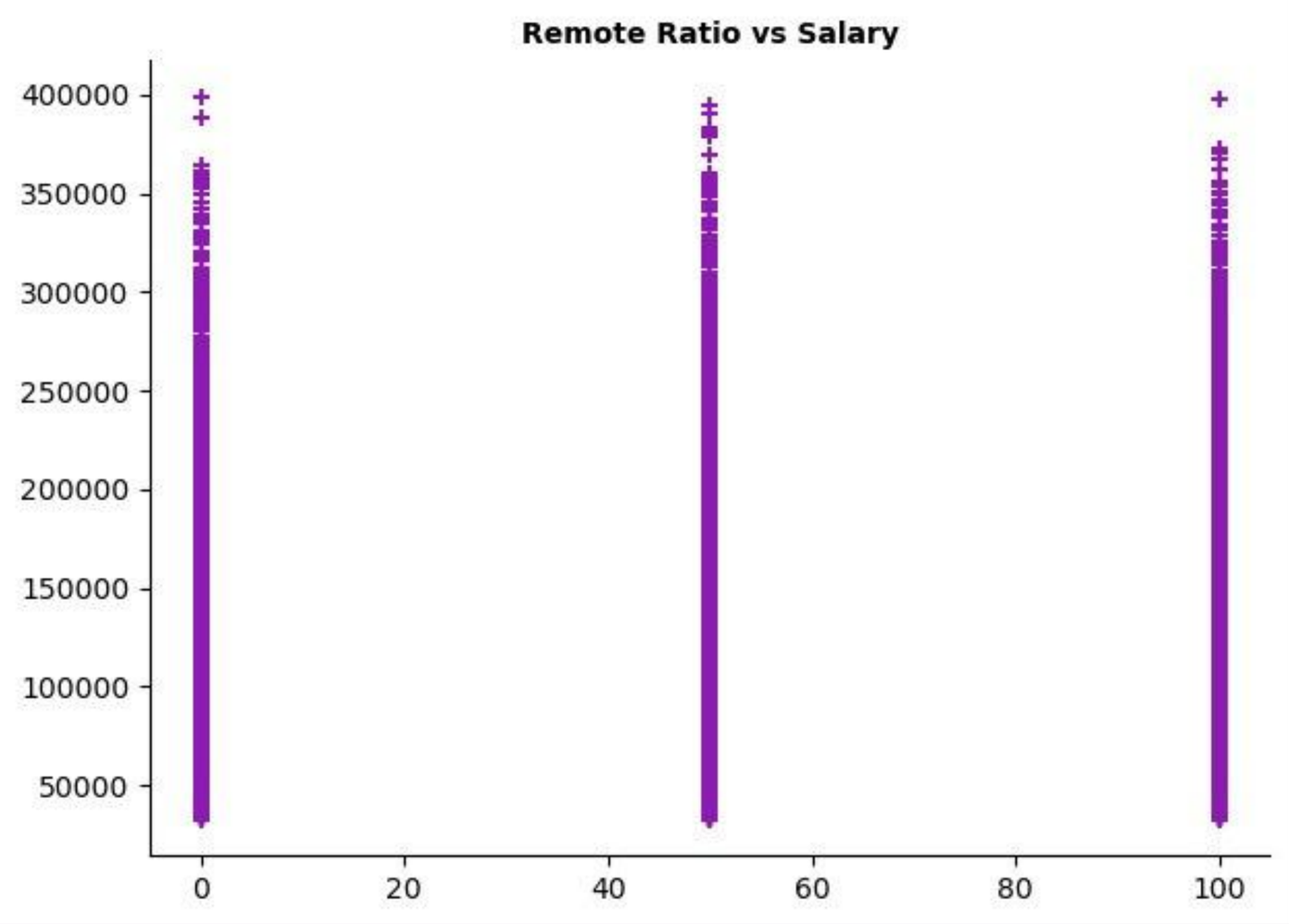
Benefits score slightly influences salary levels

Objective 3: Salary Relationships

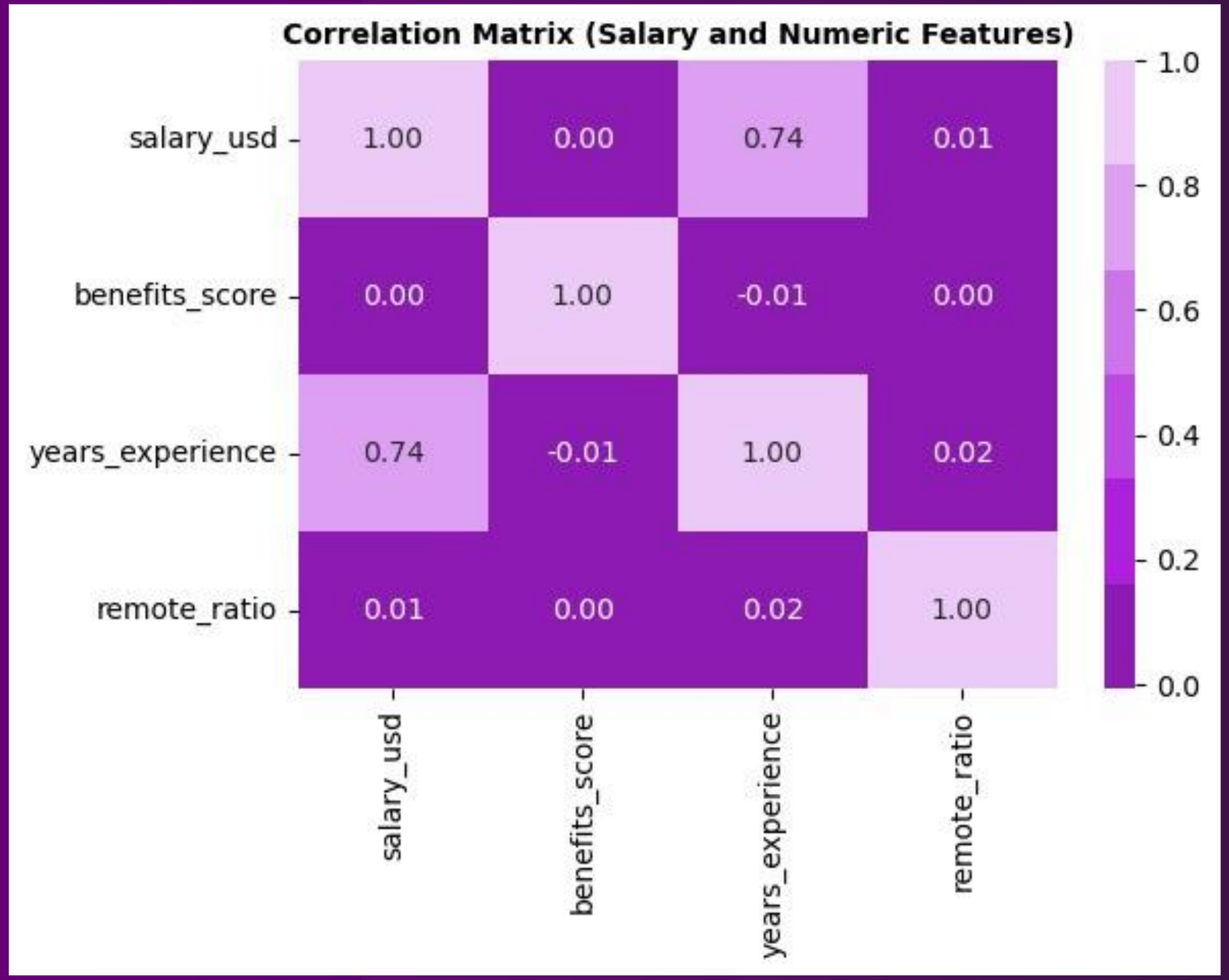


Years of experience shows moderate positive correlation with salary

Objective 3: Salary Relationships

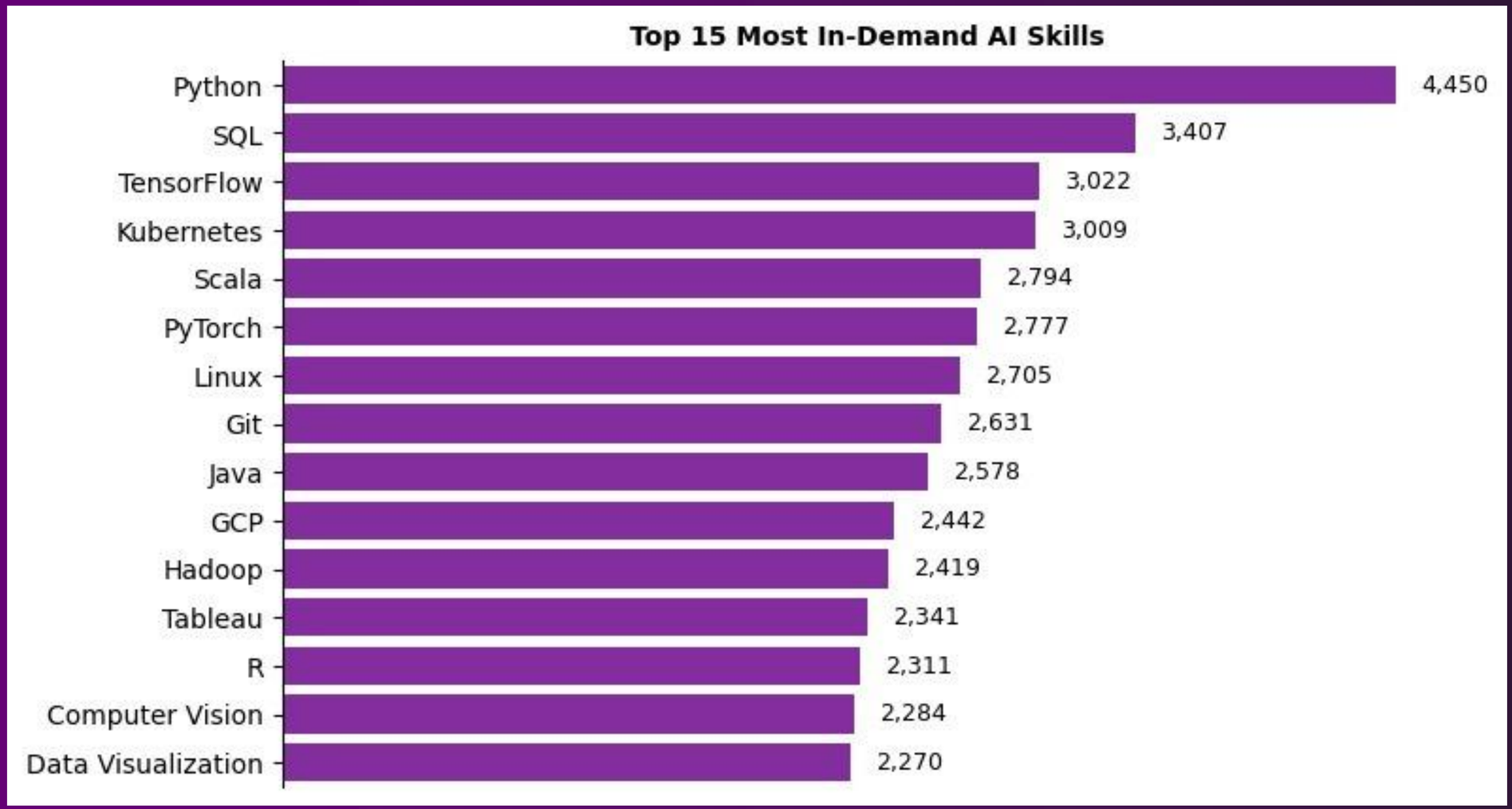


Remote ratio has weak correlation, showing that remote jobs can pay competitively



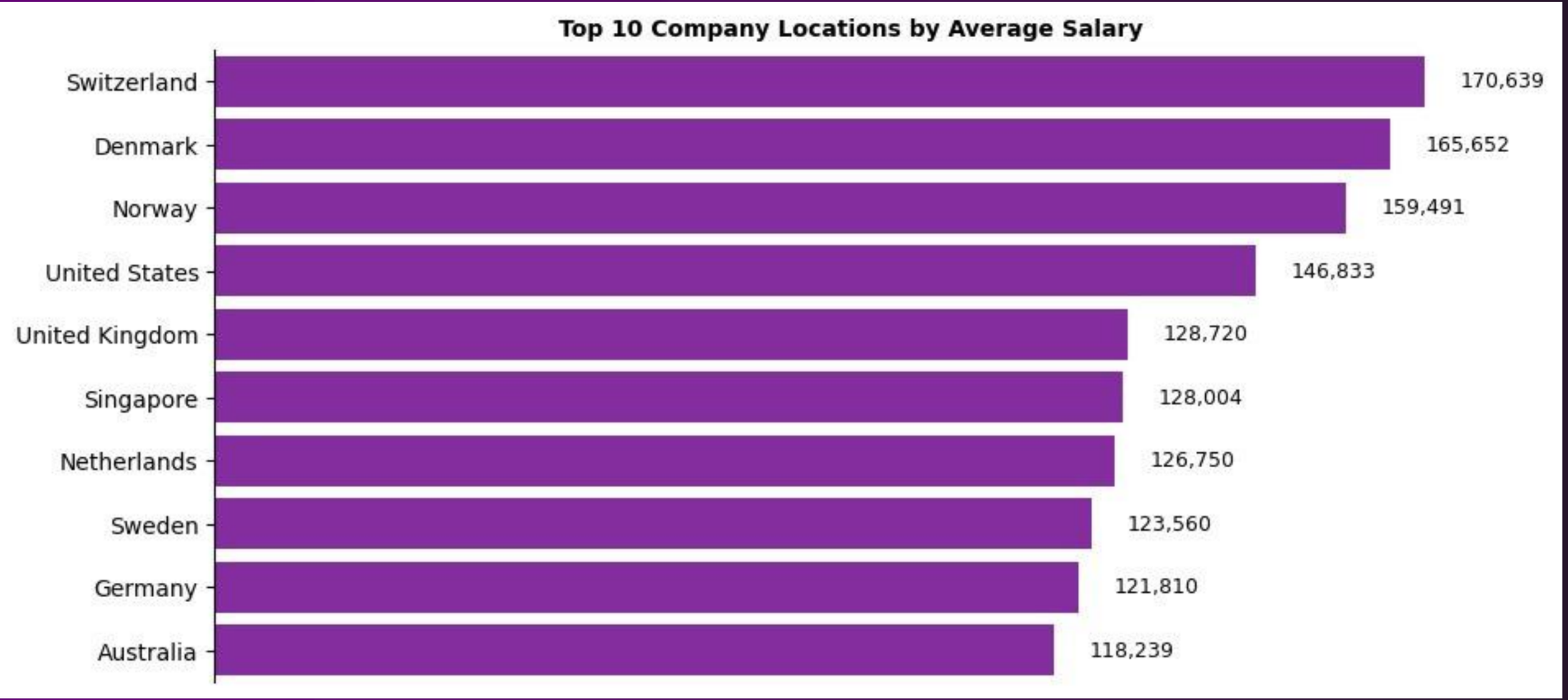
Summary of Salary Relationships

Objective 4: Most In-Demand Skills



Python, SQL, and Machine Learning appear in most AI job postings.

Objective 5: Geographical Trends



US and Western Europe dominate high-paying AI markets. Emerging hubs include Australia, Germany, and Singapore.

Predictive Analytics (Machine Learning)

Salary Forecasting Model

- Model Used: Linear Regression
- Target: salary_usd
- Predictors: experience_level, education_required, company_size, remote_ratio, benefits_score, and industry.
- Results:
- R^2 = 0.64 → model explains ~64% of salary variance
- MAE = 27,051 (average error)
- MSE = 1.3 billion

Predictive Analytics (Machine Learning)

File Edit View Run Kernel Settings Help Trusted

Markdown JupyterLab Python [conda env:base] *

[38]:

```
#Good. The model has finished learning so we now have to open our answer booklet and mark h

y_pred = model.predict(X_test)
```

[39]:

```
#Good. We are done marking, so we have to use our evaluation metrics to see whether our mod

print('R2 Score:', r2_score(y_test,y_pred))
print('Mean Absolute Error:', mean_absolute_error(y_test,y_pred))
print('Mean Squared Error:', mean_squared_error(y_test,y_pred))
```

```
R2 Score: 0.6431731597260727
Mean Absolute Error: 27051.041945979745
Mean Squared Error: 1301414743.5794754
```

Interpretation: The model performs reasonably well for estimating salary ranges.

Prescriptive Analytics (Recommendations)

For Job Seekers:

- Learn in-demand skills like Python, SQL and Machine Learning, and pair them with the experience level, role and industry you're targeting.
- Apply for jobs in bigger companies or senior roles because they usually offer higher salaries and more career growth.

Prescriptive Analytics (Recommendations)

For Employers:

- Check how much other companies pay for similar roles in the region so your salaries remain attractive to skilled professionals.
- Provide good benefits and flexible working options (like remote work). These make jobs more appealing to AI and tech experts.
- Invest in employee training to help your current staff build AI and data skills instead of only hiring new talent.

Prescriptive Analytics (Recommendations)

For Policymakers and Universities:

- Create more programs and courses in AI, data science, and digital skills to prepare students for the growing AI job market.
- Support partnerships between universities and tech companies so students can gain hands-on experience through internships or projects.
- Encourage remote and international job opportunities to connect local talent with global employers and boost digital employment.

Takehome Points

1. In this dataset, salary patterns vary with experience, company size, industry and location.

2. Python, SQL and Machine Learning appear most often in job postings. That shows demand for these skills, but it doesn't show they earn higher pay on their own.

3. A skill's value depends on context: who has it, the role, the industry, the region and the level of experience. Job seekers, employers and educators should look at those conditions, not only at skill lists.



Conclusion

This analysis of 15,000 AI job records shows that salaries vary with experience, company size, industry and location, and that Python, SQL and Machine Learning appear most often in job postings. Frequency alone doesn't make a skill valuable. Its worth depends on who has it, the role, the industry, the region and the level of experience.

Even this salary model, which explains about 64% of the variance, is best used to estimate ranges rather than exact figures. The better question is not "What skill should I learn?" but "Under what conditions does a skill become valuable?"



Thank You

FOR YOUR ATTENTION